XINLU LIU

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EDUCATION

Fudan University Sep 2021 - Jan 2024 (Expected)
Laboratory of Audio and Music Technology
M.Eng of *Computer Science*; GPA: 3.41/4.0

Huazhong University of Science and Technology Sep 2017 - Jun 2021
B.Eng in *Automation*; Percentage: 86.59/100

RESEARCH EXPERIENCE

Online beat and downbeat tracking Sep 2022 - Dec 2022
Literature research, Model design, Coding, Article writing *Shanghai, China*

- Based on the structural characteristics of music, an online beat and downbeat tracking framework with a "tilted pyramid" structure is designed. The framework improves upon the causal dilated model by introducing a small amount of latency at each layer, which significantly improves the results of real-time beat and downbeat tracking. The F-measure results for beat and downbeat are approximately 8% and 15% higher than those of the previous SOTA model, respectively.

Music source separation Sep 2022 - Dec 2022
Literature research, Coding, visualization, Musical interpretation, Article writing *Shanghai, China*

- A multi-scale Stripe-Transformer model is proposed in this study for music source separation. This model is capable of extracting stripe features for harmonics and drum beats in the spectrogram with fewer parameters while achieving SOTA performance.

Beat and downbeat tracking Jan 2023 - March 2023
Literature research, Dataset collection, Article writing *Shanghai, China*

- A hybrid network *XBeat* is proposed for beat and downbeat recognition. The *XBeat* network combines the characteristics of CNN and Transformer through feature exchange units, and introduces a novel data augmentation method. Experimental results demonstrate that *XBeat* can outperform traditional TCN baselines on most available public datasets.

Vocal register recognition Aug 2019 - Dec 2020
data collection and processing, model design, GUI design *Wuhan, China*

- A dataset for vocal register recognition is recorded and processed. A one-dimensional convolutional neural network (1D-CNN) is designed, which takes the Mel-spectrogram as input and achieved an accuracy of 99.15% on the self-constructed dataset, surpassing the performance of traditional machine learning models such as SVM.

PUBLICATIONS

Xinlu Liu, Jiale Qian, Qiqi He, Yi Yu and Wei Li: "LC-Beating: An Online System for Beat and Downbeat Tracking using Latency-Controlled Mechanism." ICME 2023

Jiale Qian, **Xinlu Liu**, Yi Yu and Wei Li: "Stripe-Transformer: deep stripe feature learning for music source separation." EURASIP JASMP

Yuxuan Wu, Yifan He, **Xinlu Liu**, Yi Wang and Roger Dannenberg: "TransPlayer: Timbre Style Transfer with Flexible Timbre Control." ICASSP 2023

Yifei Wu, **Xinlu Liu**, Xu Gan and Wei Li: "XBeat: A hybrid CNN-Transformer model for beat and downbeat tracking." CSMT 2023

Xu Gan, Yifei Wu, **Xinlu Liu**, Duan Ruilei and Wei Li: "DiffTimb: diffusion models for many-to-many timbre transfer." CSMT 2023.

COURSEWORKS

Postgraduate coursework: Audio Information Processing, Neural Networks and Deep Learning, Fundamentals and Applications of Generative Models, Machine Learning
Undergraduate coursework: Digital Signal Processing, Data Structure and Algorithm

SKILLS

Computer language	Python, C++, C, Matlab
Instruments	Guitar, Bass, Popular singing
Musical skills	Cubase, Mixing, Harmony
Deep learning	Pytorch, Keras